

Example Traceability Matrix

Customer Needs	System Requirements	Engineering Actions	Technical Objectives
<i>N1: Improve scheduling efficiency</i>	R1: The system shall evaluate scheduling algorithms based on efficiency and fairness.	EA1: Identify and implement at least three scheduling algorithms for comparison.	TO1: Develop a framework to assess trade-offs between scheduling policies.
<i>N2: Reduce patient wait times</i>	R2: The system shall simulate patient flow under various scheduling policies.	EA2: Develop and validate a discrete event simulation model of patient flow.	TO2: Provide decision-support insights into optimal scheduling strategies.
<i>N3: Optimize resource allocation</i>	R3: The system shall analyze historical demand data to identify peak congestion periods.	EA3: Perform statistical analysis of historical patient demand data to detect patterns.	TO3: Identify key factors contributing to patient congestion and delays.

Needs	System Requirements	Engineering Actions	Technical Objectives
N1	R1, R2	EA1, EA3	TO1, TO2
N2	R2	EA2, EA3	TO1, TO2, TO3
N3	R1, R2, R3	EA1, EA2, EA3	TO1, TO3